Lagrangian multiforms, 3d mixed BF theory, and Hitchin systems

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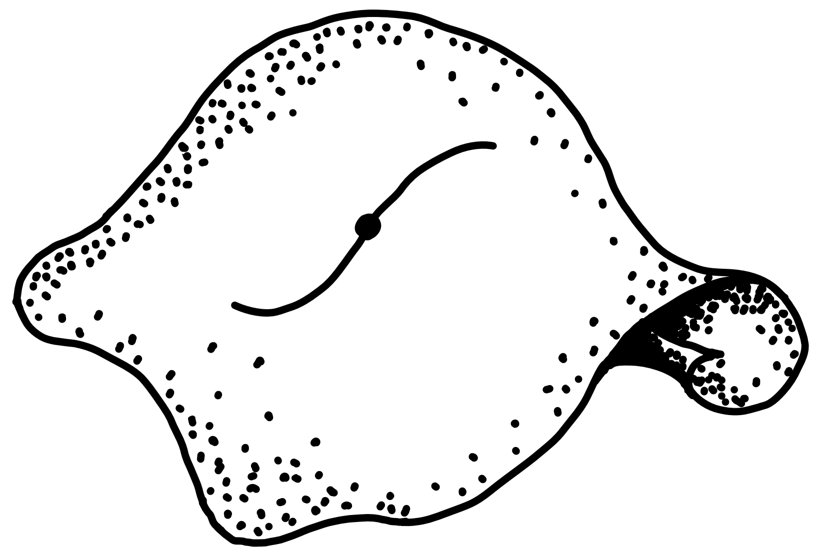

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The 3d mixed BF Lagrangian 1-form: A variational formulation of Hitchin's integrable system

Vincent Caudrelier, Derek Harland, AAS, Benoît Vicedo

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Hitchin systems

a class of integrable systems associated with the **moduli space** of **stable holomorphic vector bundles** over **Riemann surfaces** introduced in [Hitchin '87]

obtained via Hamiltonian reduction of the **cotangent bundle** to the **space of type (0,1)-fields** on a Riemann surface C

the construction and its generalisations encompass **hierarchies** of a very large class of classical integrable models

[Hitchin '87] Stable bundles and integrable systems

Lagrangian multiforms for finite-dimensional integrable systems

Hamiltonians : the traditional approach to integrability

A $2n$-dimensional Hamiltonian system is (Liouville) integrable if it possesses n independent conserved quantities H_i in Poisson involution, that is,

$$\{H_i, H_j\} = 0 .$$

One of the H_i can be taken as the Hamiltonian of interest H.

This gives us the notion of an **integrable hierarchy**: each H_i can be used define a dynamical system each with respect to a "time" variable t^i.

all the time flows imposed simultaneously on the phase spac

Hamiltonians : the traditional approach to integrability

We have a hierarchy of commuting Hamiltonian flows:

$$\underbrace{\{H_i, H_j\} = 0}_{\substack{\text{Poisson involutivity} \\ \text{of Hamiltonians}}} \Rightarrow \underbrace{\left[\frac{\partial}{\partial t^i}, \frac{\partial}{\partial t^j}\right] = 0}_{\substack{\text{commutativity of} \\ \text{vector fields}}} .$$

Solutions are functions of a **multi-time** $(t^1, \dots, t^n)$.
This implies path-independence in the multi-time.

Think not of a single integrable system,
but of the entire hierarchy it lives in.

! Takeaway message

But how would one capture integrable hierarchies
in the Lagrangian formalism?

Lagrangian multiforms

A variational framework for integrability was introduced in [Lobb-Nijhoff '09].

We replace the traditional action

$$S[q] = \int_a^b L[q]\, dt$$

by a generalised action for a collection of Lagrangians $\mathcal{L}_k$ assembled into a one-form $\mathcal{L}[q]$:

$$S[q, \Gamma] = \int_\Gamma \mathcal{L}[q] = \int_\Gamma \mathcal{L}_1[q]\, dt^1 + \ldots + \mathcal{L}_N[q]\, dt^N$$

where Γ is a curve in the multi-time $\mathbb{R}^N$ with coordinates $t^1, \ldots, t^N$.

Here q denotes generic configuration coordinates. By $\mathcal{L}[q]$ and $\mathcal{L}_k[q]$, we mean that these quantities depend on q and a finite number of derivatives of q with respect to the times $t_1, \ldots, t_N$.

Lagrangian multiforms

Applying a generalised variational principle for arbitrary curves Γ,

$$\delta_q S[q,\Gamma] = 0 \qquad \text{for all } \Gamma,$$

produces multi-time Euler-Lagrange equations:

$$\frac{\partial \mathcal{L}_k}{\partial q} - \partial_{t^k} \frac{\partial \mathcal{L}_k}{\partial q_{t^k}} = 0,$$

standard Euler-Lagrange equation for each $\mathcal{L}_k$

New (corner) Euler-Lagrange equations

$$\frac{\partial \mathcal{L}_k}{\partial q_{t^l}} = 0, \qquad l \neq k,$$

Lagrangian coefficient $\mathcal{L}_k$ cannot depend on velocities q_{t^l} for $l \neq k$

$$\frac{\partial \mathcal{L}_k}{\partial q_{t^k}} = \frac{\partial \mathcal{L}_l}{\partial q_{t^l}}, \quad k,l = 1,\dots,N.$$

conjugate momentum to q is the same with respect to all times t^k

Lagrangian multiforms

We impose an additional requirement:

integrability criterion

On the solutions of the multi-time Euler-Lagrange equations, the value of $S[q,\Gamma]$ is stationary with respect to variations of Γ, that is,

$$S[q,\Gamma] = S[q,\Gamma']$$

for all curves Γ, Γ' in the multi-time.

This implies the closure relation

$$d\mathcal{L}[q]=0 \iff \partial_{t^k}\mathcal{L}_j - \partial_{t^j}\mathcal{L}_k = 0$$

equivalent to the Poisson involutivity of Hamiltonians

on shell.

How would one describe Hitchin systems associated with Riemann surfaces of arbitrary genus variationally?

3d mixed BF theory and integrable hierarchies

Gauge-theoretic origins of integrable models

4d mixed holomorphic-topological Chern-Simons theory + defects

$\downarrow$ gauge fixing + reduction

Integrable spin chains [Costello-Witten-Yamazaki '17, '18]
2d integrable field theories [Costello-Yamazaki '19]

3d mixed holomorphic-topological BF theory + defects

$\downarrow$ gauge fixing + reduction

Rational Gaudin model [Vicedo-Winstone '22]

3d mixed BF theory

$$S_\Gamma[B,A] = \frac{1}{2\pi i}\int_{C\times\Gamma} \langle B, F(A)\rangle \quad \text{on} \quad C\times\mathbb{R}^n$$

$\mathfrak{g}$-valued $(1,0)$-form B on $C\times\mathbb{R}^n$

curvature $F(A) = dA + A\wedge A$
for $\mathfrak{g}$-valued connection
1-form A on $C\times\mathbb{R}^n$

Compact Riemann surface C of genus g
with marked points p_α, $\alpha\in\{1,\dots,N\}$

Complex semisimple Lie group G
Its Lie algebra $\mathfrak{g}$ with Lie bracket $[\cdot,\cdot]$
equipped with a nondegenerate invariant
symmetric bilinear form $\langle\cdot,\cdot\rangle : \mathfrak{g}\times\mathfrak{g}\longrightarrow\mathbb{C}$

Adding defects

$S_{BF\text{-}def}$

$$S_\Gamma[B, A, \{\varphi_\alpha\}_{\alpha=1}^N, t] = \frac{-1}{2\pi i}\int_{C\times\Gamma} \langle B, F(A)\rangle$$

fixed non-dynamical elements $\Lambda_\alpha \in \mathfrak{g}$

$$+ \sum_{\alpha=1}^{N} \int_0^1 \langle \Lambda_\alpha, \varphi_\alpha^{-1}(\partial_{u^j} + A_j(p_\alpha))\varphi_\alpha\rangle \frac{du^j}{ds}\, ds$$

type A defects at marked points $p_\alpha \in C$

G-valued field φ_α

$$-\int_0^1 H_i \frac{dt^i}{ds}\, ds$$

type B defects constructed out of polynomials invariant under the coadjoint action of G

Unifying Lagrangian one-form
[Caudrelier - Harland - Singh - Vicedo, '26]

$$S_{BF\text{-}def} \xrightarrow[\text{reduction}]{\text{gauge fixing} + } S_H = \int_0^1 \Gamma^* \mathcal{L}_H$$

transition function on C

$$\mathcal{L}_H = -\frac{1}{2\pi i} \int_{C_p} \langle L, d_{\mathbb{R}^n} \gamma \gamma^{-1} \rangle + \sum_{\alpha=1}^{N} (\Lambda_\alpha, \varphi_\alpha^{-1} d_{\mathbb{R}^n} \varphi_\alpha) - H_i d_{\mathbb{R}^n} t^i$$

Lax matrix ((1,0)-form B after reduction).

Lagrangian one-form for Hitchin systems associated with Riemann surfaces of arbitrary genus

Examples

[Caudrelier-Harland-Singh-Vicedo, '26]

Genus zero // Rational Gaudin hierarchy

$$\mathcal{L}_{RG} = \sum_{\alpha=1}^{N} \left(\Lambda_\alpha, \varphi_\alpha^{-1} d_{\mathbb{R}^n} \varphi_\alpha \right) - H_i \, d_{\mathbb{R}^n} t^i$$

Genus one // Elliptic Gaudin hierarchy

$$\mathcal{L}_{EG} = \sum_{\mu=1}^{n} p_\mu \, dq^\mu + \sum_{\alpha=1}^{N} \left(\Lambda_\alpha, \varphi_\alpha^{-1} d_{\mathbb{R}^n} \varphi_\alpha \right) - H_i \, d_{\mathbb{R}^n} t^i$$

fix $N=1$ (one marked point)

Genus one // Elliptic spin Calogero-Moser hierarchy

$$\mathcal{L}_{EG} = \sum_{\mu=1}^{n} p_\mu \, dq^\mu + \left(\Lambda_a, \varphi_\alpha^{-1} d_{\mathbb{R}^n} \varphi_\alpha \right) - H_i \, d_{\mathbb{R}^n} t^i$$

Things I didn't talk about* (but should have)

[Caudrelier-Harland-Singh-Vicedo, '26]

- 3d mixed BF theory from the reduction of a theory on the cotangent bundle to the space of holomorphic structures on a principal bundle over a compact Riemann surface

- 3d mixed BF multiform + type B defects = Hitchin system
  genus > 1

* so check out
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Open problems and future directions

- Generalised Gaudin hierarchies from 3d mixed BF multiform
  - higher order poles
  - cyclotomic generalisation
- A Lagrangian multiform generalisation of the 4d mixed Chern-Simons story
  - Lagrangian multiforms for integrable σ-models
- Classical r-matrices from gauge fixing in the 3d mixed BF setup
  - connections with algebraic approaches
- Path integral quantisation of integrable hierarchies using Lagrangian multiforms ?

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Thank you!